

Wildfires in Paraguay: Environmental and Human Impacts



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1 Introduction

Climate change already observed in Paraguay, drought events and biomass burning in the Amazon region increase the risks of forest fires in the region (de Oliveira Alves et al 2015). The greenhouse gas (GHG) emissions into the atmosphere alter the climate variables precipitation and temperature that will likely cause more severe droughts; the changes produce conditions that may increase the likelihood of wildfires. In turn, increased forest fires produced release large amounts of GHG into the atmosphere, which accelerates global warming (Nobre et al. 2016). The fires cause large-scale forest mortality in drier places like the Chaco (Brando et al. 2014) and also expand to humid areas as climate and land-use change (Le Page et al. 2017). The geographic distribution of air pollution during the periods of forest fires was analysed with data from the Ozone Monitoring Instrument (OMI/NASA) and the HYSPLIT model from the National Oceanic and Atmosphere Agency (NOAA). High average concentrations of toxic nitrogen dioxide (NO₂) were found over various regions of Paraguay, mainly in August, September and October (Recalde and Coronel 2019).

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A better understanding of the fire regimes in the country could help direct efforts to increase the capacity in the region to prevent accidental forest fires and their negative impacts on ecosystems, socio-economic well-being, biodiversity and health (Nepstad et al. 2008). A system of delicate and deep interconnections between the intense evaporation of the warm waters of the northern tropical Atlantic Ocean (ZCAS), the trade winds, the Amazon forests and the Andes mountain range, cause rainfall to reach the la Plata Basin. However, due to deforestation in dry months, forests would cease to perform their functions as a climate regulator (Marengo et al. 2007). The forest fire season in the Amazon rainforest in 2019 saw an increase in fires in the Amazon rainforest and the Amazon biome within Brazil, Bolivia, Paraguay and Peru during that year's Amazon tropical dry season (Wikipedia 2019).

Fire control is a modelling element for all forest ecosystems on Earth, and the humans use as a cleaning agent to create open land that allows agriculture. In Paraguay, during the second half of the 20th century, this practice was encouraged by an agrarian statute that sought to expand the agricultural area and the non-productive occupation of forest lands. This practice is currently prohibited by law.

Fire is indirectly responsible for desertification processes that are causing social, economic and political disorders in the planet. The evolution of forest fires depends mainly on how they originate, which can be caused by the action of atmospheric phenomena, or accidental or intentional anthropogenic origin. Due to the damaging effects of fire, preventing them is a priority, so more regional and national investigations should be conducted.

Paraguay is a landlocked country located in the Rio de la Plata basin, in the centre of South America. It has an area of 406,752 km², between parallels $-19^{\circ}18'$ and $-27^{\circ}03'$ latitude, and meridians $-54^{\circ}15'$ and $-62^{\circ}38'$ longitudes. Forest fires are considered a risk in Paraguay, are cyclical and recurrent, increasing in intensity from July to October annually (Fig. 1). The environmental vulnerability to these fires is related to the existing intrinsic predisposition of the climate and climate scenarios in the country, which are reflected in human losses or socio-economic damage. In recent times, there has been a more significant increase in the activity of forest fires globally and the countries of the region (Moreno et al. 2020), the number of outbreaks of fires, duration, extent and severity has increased. Ecosystems vulnerable to fire are increasing, and intense droughts prolong devastating fires, the Paraguayan Chaco is going through one of the worst droughts in 50 years (La Nación newspaper 2020).

The most vulnerable by air pollution caused by forest fires are children, the elderly and pregnant women in urban and rural sectors far from the sources of fire, observed by the HYSPLIT model (Stein et al. 2015), developed by the Air Resources Laboratory of NOAA (Crinó et al. 2014). During droughts, the fire season lengthens and the number of days with fire danger increases in the agricultural-livestock production areas. Better knowledge about fires will increase resilience in the regions allowing the population and resources to be better protected. The fires that occurred in the Paraguayan Chaco from August to September 2019 drew attention to its possible contamination effects throughout the country. The General Directorate of Health Surveillance activated the epidemiological alert

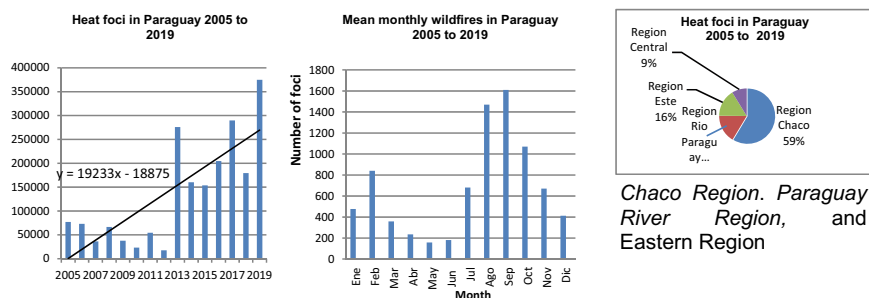


Fig. 1 The graphs show wildfire outbreak trends, months with the most fires, and the regions with the most fires, during the years 2005 to 2019. *Source* Own elaboration with data from the INPE

of risks and damage to health as a result of forest fires registered at the country level (La Nación newspaper 2019). Forest fires are among the main environmental and development problems, mainly due to their effects on health and climate change (Vázquez et al. 2007)

Climate change and forest fires in recent times have gained global interest, going from impacts to adaptation, which leads to new challenges to education and research. The purpose of this work is to present the geographic and temporal evolution of forest fires in the country and to create awareness for adaptation in the national and regional community from an interdisciplinary point of view. “Forest Fires in Paraguay: Environmental and Human Impacts” is a collaborative effort of an academic group from across the spectrum of natural sciences focused on improving the vision of forest fires for education and resilience. Exposure to air pollution is a significant cause of hospital admissions due to respiratory diseases (Machin et al. 2019). This interdisciplinary study on forest fires, climate threats, environment, the health of populations and sustainability that alters the well-being of the population, synthesises knowledge and suggests possible solutions, including stakeholder commitment. For the success of this work, national and civil organisations must play an essential role in controlling forest fires, such as putting more budgetary pressure on national governments, taking care of and managing donations appropriately. Promote the growing global consensus that the problem of forest fires be included in budget planning and proactive preparation for vulnerability reduction. The success of these wishes will depend on local community mobilisation and the connection between governments and citizens, especially in alliance with higher education institutions to influence.

One desired outcome of this study is for national and civil organisations to play a more prominent role in reducing the risk of disasters from forest fires. Several factors motivate the participation of civil society and other stakeholders, one factor is increased budgetary pressure on governments, partly due to a change in donor funding that predisposes them to share the cost of risks. Another factor is the growing global consensus that the problem of forest fires must include planning and proactive preparation and vulnerability reduction. Developing roles include

community and local mobilisation and the connection between governments and citizens, especially in alliance with higher education institutions in order to influence national government policy. The knowledge and results generated from the investigations in the universities should reach the institutions in charge of risk management and those responsible for policy formulation. This knowledge can also help introduce innovative local governance practices, in particular by trying to link research, adaptation to climate change and the provision of development services.

2 Methods, Objectives, and Organisation

This chapter is based on a non-systematic literature review of the understanding of vulnerabilities in critical sectors related to forest fires.

For a description of the 2019 wildfire season in Paraguay, up-to-date information was found on the newspapers.

The temporal evolution of hotspots is investigated using data available in “Banco de Datos de Queimadas” (Forest fires database) from the Brazilian Space Research Institute (INPE).

The study responds to the need to provide information and data in a synthesised manner to the public and to government officials to develop adaptation actions and plans, prioritising these actions according to national and regional needs.

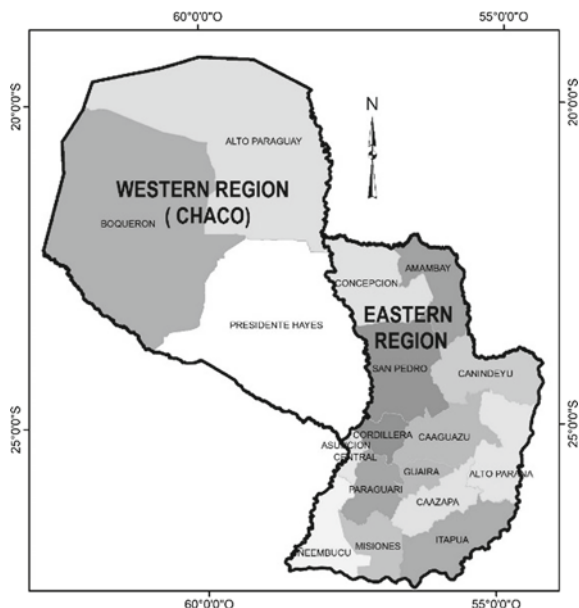
The chapter aims to provide an integrated vision of forest fires in the Western (Chaco) and Eastern (Humid) regions of Paraguay, to provide information on the climatic situations to which the population could be exposed and to what degree we can expect changes in future climate scenarios.

The chapter is organised as follows: Introduction, Methods objectives and organisation. Characterisation of the most fire region its geography and economic and productive aspects, Ecosystems function and service in the region, climate and scenarios, and trends in Paraguay. The vulnerability of the agriculture, livestock sector to wildfire. Region resilience, Summary: priority planning and preparation actions for adaptation to reduce vulnerability to wildfires in Paraguay. Conclusions. References.

3 Characterisation of the Region Where Most Wildfires Occur, Its Geography, and Economic and Productive Aspects

Wildfire outbreak data is from the Brazilian Institute for Space Research (INPE/ Burn database) that processing more than 200 images per day to detect vegetation fires. Figure 1 records the fires in the regions of Paraguay, the months with the most outbreaks and trends (INPE 2020), Fig. 1.

Fig. 2 The regions of Paraguay; Western region (Chaco) and the humid Eastern region



During the winter of 2019, a large number of heat sources were detected in the country that alarmed the citizens. In Paraguay, there are two very different scenarios for forest fires, which differ from each other geographically, environmentally and socioeconomically (Fig. 2).

The Paraguayan Chaco that includes the Departments of Alto Paraguay, Boquerón and Presidente Hayes, some 59% of the focuses of fires of the years 2005-2019 were concentrated in the Western region of the country. The Paraguayan Pantanal is located in this region and reported the first sources of fire in the week of August 14, 2019. These ecosystems have cyclical fires and, despite the vast affected area, the damage is expected to be less than in other ecosystems due to their resistance to fire. In contrast, the other region has fewer ecoregions, such as the Chaco Cerrado that were severely affected as of mid-August. Fire outbreaks in the northern part of the country were aggravated by adverse weather conditions (high temperatures, winds of more than 20 km/h and low humidity); The National Emergency Secretariat (Secretaria de Emergencia Nacional 2020) reported that about 325,000 ha were affected by the fires. Northern Chaco and the rest of the territory showed patterns associated with heat sources and the change in land use.

Additionally, due to strong winds from the north, fires that originated across the border in Bolivia entered Paraguayan territory, affecting private properties whose owners mobilised to combat them. This region has characteristics associated with extreme climatic events such as drought, high temperatures, and wind that make wildfires very difficult to combat and extinguish them due to their large extent. This Chaco fires were a major environmental catastrophe, and they called attention to the

need for a detailed assessment and the adoption of measures that drive restoration processes.

In these ecosystems the fires are cyclical, and despite the vast affected area, the damage is expected to be less than in other ecosystems due to their resistance to fire. Forest fires usually have several origins and can be caused by natural or spontaneous combustion, lightning, or generated by traditional practices of burning fields. In economic terms, these claims of intentional ignition are more relevant because of their consequences for biodiversity and productive systems. Wildfires that occur in the forests, which affects several areas of the region, have severe consequences on natural resources, as well as for the livestock sector directly and for a prolonged period.

In the Eastern Region of Paraguay, wildfire outbreaks generally affect livestock grazing and extensive agricultural areas, in Bella Vista Norte of the Amambay department there are sugar cane plantations, which regularly burn both intentionally and no intentionally. In both Eastern and Western Region, wildfires affect animal and human health due to smoke inhalation and poor air quality.

The calculation of environmental, monetary services is complicated and depends on the characteristics of each type of ecosystem, timber products and the market value of carbon sequestration, among others. In the case of forest fires, emissions from gases are a cost that local, national, regional, and the global community bear.

Forest and grassland fires had a substantial impact on the Chaco economy in 2019; forest fires not only cause damage to ecosystems and biodiversity but also have enormous social and economic impacts on the surrounding communities.

From January to August of 2019, some regions of South America were affected by increases in the number of forest fires; Brazil had 75,336; Venezuela follows with 26,491; Bolivia with 17,154; Peru with 5,681; finally Paraguay, with 9,861 wildfires.

In March–May of 2020, the total number of heat sources reached 652,917 active sources, some countries with more active sources detected with the VIIRS 375 m sensor are Venezuela with 278,266 sources, followed by Colombia 121,636, Brazil 78,548, Argentina 7,125 and in fifth place Paraguay 55,478. In Paraguay, it is not common to see so (Fig. 3) many active heat sources in March, the high peaks are observed from August to October. Such behaviour may be due to the existence of much combustible material, the performance of prescribed burns, climatic (dry), economic, social and cultural factors, among other factors (INPE 2020).

The first months of 2020 were anomalous; the President Hayes department suffered 21143 fire outbreaks, Boquerón 9057, Alto Paraguay 2961, Misiones 2482, Ñeembucú 5356, Paraguari 2482 and Cordillera 1569 outbreaks.

The occurrence of the El Niño events (October to March) in the Paraguayan Chaco has an essential influence on wildfires (Pastén 2006) with decreased precipitation in general during La Niña events. During the cold ENSO phase, this, decrease in rains contributes to prolonged droughts such as those that occurred during 1999–2000 (Grassi et al. 2005). This water deficit, in turn, causes droughts and significant reductions in water in forests. Illegal burning of fields and crops complicates this situation and increases the risk of forest fires. High risk exists from the spread of forest fires in Brazil, Bolivia, Paraguay and Peru.

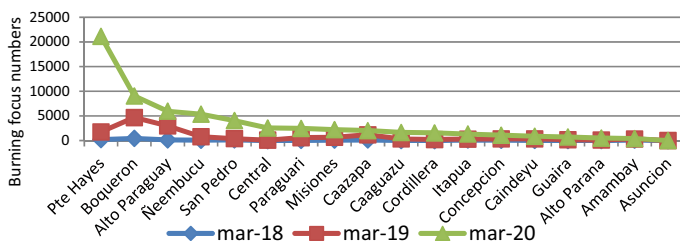


Fig. 3 The focus of burning in the regions of Paraguay, March 2018, 2019 and 2020 (NPP VIIRS 375 m INPE, 2020)

4 Ecosystems Function and Services in the Region

In urban ecosystems, rapid urbanisation turns cities into centres of environmental problems, and these urban areas are increasingly engines of innovation and socio-economic development. They must adapt to local and regional socio-environmental challenges; overpopulation makes urban areas increasingly key to sustainability. Balanced growth and development is part of the solution at local and regional scales. Therefore, it necessary to take into account the ecosystem services of cities and consider them as urban ecological systems (Leal Filho et al. 2020). Increasing knowledge would be the basis for improving practices in the management of urban ecosystems, and being able to respond to the growing need to develop urban areas and implement plans for their sustainability and resilience, which ultimately leads to the human well-being of the population.

The objective of studying urban ecosystems is to promote adoption and provide guidance for ecosystem approaches in planning and management of urban ecosystems, to improve the ecosystem services provided by urban ecosystems, and human well-being. It also aims to build a network of scientists, professionals, and policy and decision-makers to advance our ecological understanding of urban areas, to better plan and design urban ecosystems and to increase their sustainability and resilience.

Fire is a natural element for the functioning of numerous forest ecosystems; it is one of the natural elements that influence plant communities over time and plays an essential role in the health of particular ecosystems. The established dynamics between human action and fires and the higher frequency of the El Niño phenomenon have given rise to a situation in which fires are a significant threat to many forests and diversity (Nasi et al. 2001). Uncontrolled fires that originate without direct or indirect human intervention exert a significant influence on certain types of vegetation (Pritchett 1991). Forest fires can be unwarranted and uncontrolled in which large amounts of combustible material are burned and spread rapidly, often destroying everything in their path. The effects of fires are associated with the intensity, recurrence, and duration of a forest fire. The impacts of fires on a forest

ecosystem involve alterations of the three main components: soil, water, vegetation and fauna,

Chile's wildfires affect the country on various fronts, generating three main types of problem: Environmental problems, Social problems, Economic problems (Ubeda and Sarricolea 2016) pointed out that fires cause soil degradation, mainly after the event, the level of degradation will depend on the topography of the site, the intensity of the erosive process, the regeneration rate of the post-fire vegetation cover, the recurrence, intensity and duration of the fire (Caon et al. 2014). Gonzalez (2017) pointed out that the main impacts occur at the level of the physical, chemical and biological, and soil productivity (Garcia-Orenes et al. 2017). Wildfires may produce several changes in the short and long term in the landscape and the soil system (Zavala et al. 2014). After fires, the nutrient content increases in the first ten centimetres of the surface, due to ash deposition, mineralisation of nutrients and formation of stable structures. This process was the basis of indigenous slash and burn agriculture and forest fires can modify the hydrological cycle at the level of the river basin. The climatic condition of the area is decisive on the probability of a fire occurring. Wildfire seasonality explains the fire regime, while the modification of the structure and composition of the vegetation due to forest fires can affect ecosystem services.

In forests, fire can have devastating effects on forest species of vertebrates and invertebrates, not only because it causes death, but also because it causes more lasting indirect effects such as stress and disappearance of habitats, territories, shelter and food. The disappearance of organisms of great importance for forest ecosystems, such as invertebrates, pollinators and decomposers, can significantly delay the recovery rate of the forest (Boer 1989).

Wildlife may be affected both negatively and positively. With the destruction of forests, many species of insects, mammals and birds that survive a fire lose their habitat and are forced to move. Fire plays a vital role in maintaining the health of particular ecosystems. However, in addition to climate change, human use of fire is now a threat to forests and their biodiversity; therefore, human action becomes vital (Meijaard et al. 2008). In Chaco fires almost always occur due to human intervention implies that the adverse effects of fires will be much higher than the positive effects. Despite the positive and negative effects of a fire and the discussion surrounding it, it can be generalised that fires and burns have impacts on different aspects of the environment such as air quality, wildlife, soil, and soil biota.

5 Climate and Scenarios, Extreme Events, and Climate Trends for Paraguay

One of the problems that arise at the global and local levels is the impacts of climate change on society, the economy and the environment; we use Global Climate Models to analyse this, evaluating changes in the temperature and precipitation at the regional and local level. The Regional Climate Models (R.C.M.s) reducing climate characteristics from global to regional and report in detail on a particular

region. The high resolution of the R.C.M.s better simulates the current climate in its interaction with the terrain, more closely predicts future climate change, small area representation, better simulates and predicts extreme weather events, generates more detailed data for analysis and impact studies at the local and regional levels.

Climate scenarios for Paraguay were carried out with the Eta-20 km model for South America (Chou et al. 2014; Dereczynski et al. 2020), the results indicate that the country would be affected by climate change, with the most significant increases in temperature for the period 2031–2050 in the department of Amambay (+3.1 °C). The scenario shows that rainfall will decrease in the first three decades and then increase in the last decade of the period considered. Precipitation shows a decrease during the first three decades and in the 2041–2050 decade increases slightly but very close to its typical values for both scenarios. The highest increases in temperature from 2031 to 2050 are expected in the department of Amambay (+3.1 °C) (northeast of the Eastern Region). The departments of the country that expect an increase in temperature and decrease in precipitation may be more severely affected during La Niña years, because this phenomenon produces a decrease in rainfall in much of the country. The same can occur in those regions where temperature and precipitation are expected to increase since in El Niño more precipitation than average occurs in much of the country. The climate scenarios for the Chaco were carried out by Max Pasten with the Climate Scenarios Technical Team of “Centro de Conocimiento Del Gran Chaco Americano y Cono Sur” (SERIE CLIMA 2017).

The South America Low-level Jet (SALLJ) is a climatological feature with a critical role in the spatial and temporal distribution of precipitation in the region. Previous studies focused on the mechanisms and variability of the SALLJ in the central Andes (i.e., southeast Peru, Bolivia, and Paraguay), this study show that the SALLJ in the northern branch exceeds 10 ms^{-1} during September–February and the frequency can be as high as in the central region. When the SALLJ is active, wind speed is vital in the spread of fires along with rainfall. The frequency and intensity of the SALLJ in the northern Andes have substantially increased in the last 39 years (Jones 2019).

6 The Vulnerability of the Agriculture and Livestock Sector to Wildfire

The primary division that characterises agriculture in Paraguay is one between commercial and family agriculture. Wildfire from intentional and nonintentional burning of pastures may spread to sugar cane fields and damage other row crops, including limited-resource family farms. Of recent concern is the expansion of eucalyptus plantations for biomass production (Manrique et al. 2020), and silvopastoral system—which combine tree plantations with cattle grazing in Eastern Paraguay, and the new practice of forest grazing in the Chaco Region. Under the right conditions, these systems are susceptible to wildfire and may serve as a source affecting other systems.

The agricultural sector contributes 30.4% of Gross Domestic Product and 35% when agribusiness is included. It generates 40% of national exports and occupies nearly 40% of the country's workforce (World Bank 2014). Agriculture faces many types of production risk, many of which are agroclimatic in nature, and the most important of which is drought. The available literature does not mention vulnerability to wildfire as a significant risk in row crop agriculture, and livestock production, on the other hand, is susceptible to wildfire risk. With the rise of agroforestry and forest plantations on former livestock grasslands, wildfire risk to pastures that sustain beef production is likely to rise. However, it is essential to note that the sharp distinction between livestock and forest systems is being blurred by new practices (including forest grazing), and the reclassification of palm savannahs, in the Western Region of Paraguay, that are natural fire ecosystems, as forestlands.

In Paraguay, the bulk of agricultural production is concentrated in three crops, soy, maize and wheat. The gross value of the production of these three products represents about 76% of total agriculture in Paraguay, with soybeans alone accounting for 46%. Irrigated rice is also growing in importance at the commercial scale. Occupying a smaller area, but not less important from the socio-economic point of view for family farming, are cassava, beans, white corn, cotton, sesame, vegetables and fruit trees. Family farming mainly using family labour and is composed of production units under 50 ha in size in the Eastern Region and 500 ha in the Western or Chaco Region; family farmers represent 91% of all the farms. The geographical distribution of family farming is very heterogeneous; however, it is mainly concentrated in the departments of San Pedro, Caaguazú, Caazapá, Paraguari, Guairá and Cordillera, in the Eastern Region of the country. In general, they coexist in the same areas as commercial holdings. The area cultivated by family farming has remained constant or decreased for some products, during the last decade (SENACSA 2019). Some crops on family farms, especially sugar cane, are affected by local wildfires in small 2–10 ha patches that may appear unimportant even in aggregate but can be devastating to a family farm with low capital and low technology.

Currently, more than 123,000 livestock producers exist with a stock of nearly 14 million head of cattle (Ministerio de Hacienda 2018). Today, livestock is a crucial sector of the Paraguayan economy, especially for its contribution to exports (beef), which have increased sharply, reaching 1.5 billion USD in 2018 (Ministerio de Agricultura y Ganadería 2013). Many traditional ranchers still burn their natural grasslands in the lower (humid Chaco) and the natural grasslands of the Oriental Region in the austral winter months. When drought, high temperatures and high winds create unfavourable conditions, they can spread over broad areas, and usually do so every year. Many ranchers burn only small fields after first rains and with favourable or no wind using controlled backfires. In the northern or dry Chaco, burning pastures is not customary. A law regulates wildfire use for this purpose, Law 4014/2010 that regulates this activity. However, the Ministerio Del Ambiente y Desarrollo Sostenible (MADES) has little human and financial resources to enforce it.

The expansion of soybeans was mainly at the expense of forest areas and pastures. Currently it also expands on lands traditionally occupied by family agriculture, leased or sold by peasant farmers, mainly in the departments of Caaguazú, San Pedro and Caazapá (Fraisie et al. 2008). Commercial agriculture and livestock are growing in the area, whereas family farming is stagnant or decreasing. Risk evaluations focus on the value chains of soybeans, corn, wheat, rice, cassava, sesame, cotton, sugar cane and vegetables, in addition to livestock.

Agroclimatic risks, along with pests and diseases, affect grain and oilseed production in Paraguay. The main risk factor for soybeans is summer drought (mainly in January), whose impact is exacerbated by high temperatures and enhanced in producing areas with soils with less water retention capacity (more sandy soils in San Pedro and Canindeyú). Weeds and other pests and diseases systematically affect grain production in Paraguay. Their intensity varies depending on climate conditions and crop management. La Niña years usually exacerbate seasonal drought and are responsible for losses as high as 38% (World Bank 2014). In the case of second planting (zafriña), maize, early frosts, and late summer drought are the leading causes of yield variations. The occurrence of early frosts may threaten and limit the sowing of the second maize if the soybean harvest falls behind. Finally, excess rainfall in the harvest season of the second maize (June) is frequent at harvest time.

For livestock production, drought is a significant risk. Drought occurs in all regions of Paraguay, affecting the Western region or Chaco more drastically from June to September. In this period of the year, rainfall is scarce, and wildfires tend to occur. Droughts occur at apparent semi-cyclical intervals of 4–5 years, with extreme periods every ten years (Pezzoli et al. 2016). Production losses include reductions in pregnancy rate, 30%; rate birth rate, 25%; weight loss, 20%; and decrease in slaughter rate, 10%. Wildfires effects are immediate, as pasture burns along with forest reserves, and hay. Cattle often need to be physically transported to rental properties or sent to slaughter underweight.

In Paraguay, prolonged drought, heat and the burning of grasslands aided by strong winds, spread flames, giving rise to intense forest fires on reserves and commercial ranches. Since wetlands and streams are almost dry, wildlife must migrate or perish. With such a large sector of the economy facing production risk, it is imperative to assume the challenge of developing a management plan for these situations, to reduce their negative impacts (World Bank 2014; Food and Agricultural Organization 2017).

7 The Adaptive Capacity of the Region

Law N° 4014 (Fire Prevention and Control, 5/12/2009) establish suitable regulations to prevent and control rural, forest, vegetation and interface fires, in Paraguay. The law also seeks to reduce the uncontrolled burning of pastures, forests, scrublands. It forbids burning fallow, natural fields, sawdust or any other cereal, of

legumes or type of flammable organic material that could generate any of the fires defined in this law. However, the law has not been regulated to date. Besides, the law enforcement fails due to lack of resources at the municipal, departmental, and Ministry level. The country's municipalities are the law enforcement application authority for wildfire in coordination with the Paraguayan Network of Prevention, Monitoring and Fire Control, which is a specialised unit that is coordinated by the National University de Asunción, together with public and private institutions related to the matter.

Forest ecosystem management: a need for the environment (Kimmins 1974), and must serve the needs and desires of the surrounding people (Lutz et al. 2001), Paraguay's current population of around seven million puts pressure on the country's forests. New paradigms must emerge as models for forest management, and forestry faces new social pressures, ideas, and concepts, as well as the ability to anticipate their long-term and medium-term social and environmental consequences. These forests and values will need to meet the current and future needs of both the local population and society (Franklin et al. 1997, 2002; Franklin 1997).

Forest ecosystems are tree-dominated landscape areas. They consist of biologically integrated communities of plants, animals, and microbes along with local soils and atmospheres with which they interact, insects, the disease may have recently altered forests, but continue as forests due to the biological and physical legacies of the previous forest. Forest ecosystem management is the management of forest ecosystem disorder processes and regimes to sustain desired ecosystem values and services. It is also the management of the human use of the forest and its interaction with it because men are part of forest ecosystems. An understanding of structure, function, and energy, nutrient, and information flows are paramount for this type of forest management.

The environmental impact of forest fires is not limited to biodiversity, the Eastern region and the Chaco, in particular, suffered dangerous wildfires in late austral winter and spring of 2019. The fires negatively impacted biodiversity may have increased desertification in the northwestern-most regions, and air and water quality negatively affected. Humans cause between 80 and 90% of fires. Biodiversity of the burned area changes its structure and composition. Wildfire negatively affects soil and water since both are related. The vegetable mantle disappears, and with it, the natural barrier that retains water. Wildfires emit greenhouse gases such as carbon dioxide into the atmosphere. To this must be added the economic costs derived from any forest fire. The ecosystem loses its appeal for both leisure and tourism activities. To all, this must be added the economic cost of regeneration. Land-use changes cause most forest fires. However, in some ecosystems such as cerrado and savannahs, wildfires play a vital role as they have a regulatory role in ecosystems and are crucial to biodiversity.

8 Summary, Planning and Preparation Actions for Adaptation to Reduce Vulnerability to Wildfires in Paraguay

Fire management is complex and involves many stakeholders; the ability to integrate external information is key to proactive planning and preparation for wildfire mitigation. Seasonal climate forecasts complement satellite data, and local populations, agricultural and livestock firms, insurance companies, businesses, health and transportation professionals, and others have an interest in fire management for adaptation. Institutional actors hold most of the decision-making power; they can incorporate climate forecasts and climate change scenarios into participatory planning and preparation processes that are central to current risk management approaches. They should also communicate monitoring, alerts, and adaptation information to citizens ahead of time and in real-time. Stakeholders must evolve from being passive users of scientific information to being protagonists in a participatory and iterative consultation process for readiness. Interaction among public officials, scientists, and stakeholders requires stable and long-term funding and institutional incentives for preparation rather than reactive action. Interdisciplinary research agendas, long-term programs, sustained partnerships, and continuous outreach and training will be crucial to the adaptation to wildfire in a changing climate.

Some recommendations that may be of interest to local actors are the following.

- Consolidate a Valuation System for economic losses from forest fires in the entire region
- Involve and integrate the community of the Paraguayan Network for Fire Prevention and Control, under art. 4 of Law 4014/2010
- The application of fire prevention should not hinder economic development. A focus on positive effects on the environment is a better approach,
- The use of satellite monitoring techniques should increase, due to the climatic conditions of the Chaco, because of its low population density and difficult to access environment.

9 Conclusions

Paraguay has two great regions: Western region (Chaco) and the humid Eastern region divided by the river that bears his name, with three departments in Chaco and fifteen in the Eastern region. Burning in Paraguay, in the study period, showed a positive trend, given by the equation $y = 19233 \times -18875$.

The highest monthly averages in the study period are observed from July to October.

The largest area of fires occurs in El Chaco Humedo (Wet Chaco) (Western Region adjacent to the Paraguay River) where the vegetation is exuberant. Since 2005 the regions most affected by burns are (i) Chaco (59%), (ii) Eastern regions, and (iii) the adjacent areas to the Paraguay River (16% each). The region with the fewest fires is the Central department that surrounds Asuncion (Capital) with 9%.

During the first months of the year, the occurrence of fires is relatively low, except for the year 2020, in March the department of Presidente Hayes was exceptional where 21,143 outbreaks occurred, and in Boquerón with 9,057 sources of burning, as shown in Fig. 3.

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